



大车运行机构设计计算书

Calculation Sheet for Design of Crane Operation Mechanism

1. 大车运行机构

1. Trolley Travelling Mechanism

机构级别: Mechanism grade:	M 8		
	T7	L3	
大车自重（除移动载荷外）: Self-weight of trolley (except the moving load):	DL= 6.5	t	
大车重心高度（相对轨面）: Gravity center height of trolley (relative track):	$h_{gc1}= 750$	mm	（满载） (Full Load)
	$h_{gc2}= 36$	mm	（空载） (No Load)
大车运行速度: Trolley travelling speed:	$v_{yc1}= 50$	m/min	
	0.83	m/s	
许用起动时间: Allowable starting time:	$[t_q]= 4.0$	s	
许用制动时间: Allowable braking time:	$[t_z]= 4.0$	s	
许用起、制动加速度: Allowable starting and braking acceleration:	$[a_z]= 0.21$	m/s ²	
小车轨道高度: Track height of hoist trolley:	$h_b= 1336$	mm	（相对大车轨顶面） (Relative track top surface of trolley)
基距: Wheel base:	B = 3200	mm	
轨距: Wheel track:	$S_g= 6900$	mm	
最大外伸距（停车位）: Maximum outreach distance (parking spot):	$S_t= 1150$	mm	（小车运行的极限位置） (Limit position of trolley travelling)

2. 车轮和轨道

2. Wheel and Trolley Rail

（1）选型
(1) Model Selection

大车车轮数: Number of trolley wheels:	$n_{nl}= 4$	
选择车轮直径: Diameter of selected wheel:	$D_{yc}= 250$	mm
车轮型号: Wheel model:		
车轮材料:	42CrMo 锻造	

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姓名: 彭军
Name: Peng Jun
日期: 2024 年 12 月 24 日
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Wheel material: 42CrMo forging

$\sigma_b = 840$ MPa

车轮轴直径:
Diameter of wheel axle: $d_{lc} = 85$ mm

单个车轮重量:
Weight of single wheel: $m_{lc} = 76$ kg

大车轨道型号:
Model of trolley rail: 30kg/m

轨道尺寸:
Trolley rail size: $A = 107.95$ mm
 $F = 107.95$ mm
 $b = 60.33$ mm
 $r = 7.94$ mm

轨道材料:
Trolley rail material: U71Mn (GB/T2585-2007)

$\sigma_b = 880$ MPa (强度极限应大于 800MPa)
(The strength limit shall be greater than 800MPa)

(2) 踏面疲劳和静强度校核

(2) Tread Fatigue and Static Strength Check

			动态系数 Dynamic load efficient	最大轮压 Maximum wheel load
大车自重引起的轮压: Wheel load caused by the trolley's self-weight:	$P_{Lc11} = 15.94$	kN	$\phi 4$	1.38 22.00
	$P_{Lc12} = 15.94$	kN	$\phi 4$	1.38 22.00
风载荷引起的轮压: Wheel load caused by wind load:	$P_{Lc21} = 0.00$	kN	(满载) (Full Load)	
	$P_{Lc22} = 0.00$	kN	(空载) (No Load)	
由惯性载荷引起的轮压: Wheel load caused by inertial load:	$P_{Lc31} = 0.522$	kN	(满载) (Full Load)	$\phi 5$ 1.5 0.78
	$P_{Lc32} = 0.018$	kN	(空载) (No Load)	$\phi 5$ 1.5 0.03
移动载荷引起的轮压: Wheel load caused by moving load:	$P_{Lc41} = 60.90$	kN	(满载) (Full Load)	$\phi 2, \phi 4$ 86.09
(移动载荷在极限位置) (Moving load is at the limit position)	$P_{Lc42} = 12.18$	kN	$\phi 2, \phi 4$	17.22
	$P_{Lc51} = 34.95$	kN	(空载) (No Load)	$\phi 2, \phi 4$ 50.18
	$P_{Lc52} = 6.99$	kN	$\phi 2, \phi 4$	10.04
正常工作状态下的最小轮压: Minimum wheel load under the normal working status:	$P_{Lcmin} = 22.41$	kN		

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正常工作状态下的最大轮压: Maximum wheel load under the normal working status:	$P_{Lmax} = 77.37$	kN	108.87
平均轮压: Average wheel load:	$P_{Lmean} = 59.05$	kN	$\left(P_{Lmean} = \frac{P_{Lmin} + 2P_{Lmax}}{3} \right)$
与材料有关的许用线接触应力常数: Allowable line contact stress constant related with material:	$k_1 = 7.20$	N/mm ²	($\sigma_b > 800$)
车轮与轨道有效接触宽度: Effective contact width between the wheel and trolley rail:	$l = 44.5$	mm	
车轮转速: Wheel rotation speed:	$n_{yc} = 63.66$	r/min	
转速系数: Speed coefficient:	$C_1 = 0.91$		(根据车轮直径和运行速度) (Based on the wheel diameter and travelling speed)
工作级别系数: Working grade coefficient:	$C_2 = 0.80$		(机构级别 M8) (Mechanism grade: M8)
疲劳强度校核公式: Fatigue strength check formula:	$P_{Lmean} \leq k_1 D_{yt} l C_1 C_2$		
	$k_1 D_{yt} l C_1 C_2 = 58.25$	kN	
轨道顶面为圆弧形, 考虑 k_1 增加 10% The trolley rail top surface is arc-shaped increases for 10% considering k_1	64.07		
	车轮满足疲劳强度要求! The wheel meets the requirements for fatigue strength!		
静强度校核: Static strength check:	$P_{Lmax} \leq 1.9 k_1 D_{yt} l$		
	$1.9 k_1 D_{yt} l = 152.02$	kN	
	车轮满足静强度要求! The wheel meets the requirements for static strength!		

3. 轴承 3. Bearing

调心滚子轴承: Self-aligning roller bearing:	22216EAE4C3	NSK 产品 NSK product	
骨架油封: Skeleton oil seal:			
轴承支承距离: Bearing support distance:	$L_z = 204$	mm	(车轮中心到轴承中心的距离) (Distance from wheel center to bearing center)
轴承外径: Bearing outer diameter:	$D = 130$		
轴承内径: Bearing inner diameter:	$d_z = 75$	mm	
宽度: Width:	$B = 31$	mm	

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$$L_{10h} = \frac{10^6}{60n_{ye}} \left(\frac{f_t \times C_r}{P} \right)^\varepsilon, P = f_d(XF_r + YF_a)$$

$C_{r0} = 275$ kN

额定寿命: [L_{10h}] = 25000 小时
Rated life: Hour

温度系数: $f_t = 1.00$ (工作温度 < 120° C)
Temperature coefficient: (Working temperature: < 120°C)

寿命系数: $\varepsilon = 3.33$ (滚子轴承)
Life coefficient: (Roller bearing)

冲击载荷系数: $f_d = 1.20$ (中等冲击力)
Impact load efficient: (Medium impact force)

轴向载荷系数: $\gamma = 0.10$
Axial load coefficient:

轴向载荷: $F_{a1} = 0.00$ kN
Axial load:

$F_{a2} = 7.74$ kN ($F_a = \gamma P_L$)

径向载荷: $F_{r1} = 43.42$ kN
Radial load: $F_{r2} = 33.94$ kN
$$\left(F_r = \frac{P_L}{2} \pm \frac{F_a}{L_z} \cdot \frac{D_{yc}}{2} \right)$$

判断系数: $e = 0.22$
Judgment coefficient: $F_a / F_r = 0.00$ 0.22793

径向动载系数: $X = 1.00$ 0.67
Radial dynamic load coefficient:

轴向动载系数: $Y = 3.10$ 4.60
Axial dynamic load coefficient:

当量动载荷: $P_1 = 52.11$ kN
Equivalent dynamic load:

$P_2 = 70.00$ kN

$P = 58.07$ kN

额定动载荷: $C_r = 264.00$ kN
Rated dynamic load:

实际轴承寿命: $L_{10h} = 40747$ 小时
Actual bearing life: Hour
轴承满足寿命要求!
The bearing meets the life requirements!

4. 电动机

4. Electric Motor

(1) 载荷计算

(1) Load Calculation

自重载荷: $P_{Gc1} = 209.9$ kN (满载)
Self-weight load: (Full Load)

$P_{Gc2} = 149.1$ kN (空载)
(No Load)

大车水平惯性载荷: $P_{Ac1} = 4.5$ kN (满载)

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Horizontal inertial load of trolley:		(Full Load)
	$P_{Ac2} = 3.2$	kN (空载) (No Load)
车轮轴承摩擦系数: Wheel bearing friction coefficient:	$\mu_c = 0.015$	(滚动轴承) (Rolling bearing)
滚动摩擦系数: Rolling friction coefficient:	$\kappa\chi = 0.40$	(轨道为钢质、圆顶) (The trolley rail is made of steel with dome)
附加摩擦阻力系数: Additional friction resistance coefficient:	$C\phi = 1.50$	(有轮缘, 分别驱动) (They are designed with rim and driven respectively)
运行摩擦阻力: Travelling friction resistance:	$P_m = mg \frac{\mu_c d + 2k_c}{D_{yt}} C_f$	
	$P_{mc1} = 2.61$	kN (满载) (Full Load)
	$P_{mc2} = 1.86$	kN (空载) (No Load)
运行坡度阻力: Travelling grade resistance:	$P_p = mg \cdot \sin\alpha = mg \cdot m_p$	
坡道阻力系数: Slope resistance coefficient:	$m_p = 0.002$	
	$P_{pc1} = 0.42$	kN (满载) (Full Load)
	$P_{pc2} = 0.30$	kN (空载) (No Load)
运行风阻力: Travelling wind resistance:	$P_{WI} = C_{qw} A_w$	
风力系数: Wind coefficient:	$C = 1.40$	
	$P_{Wc1} = 0.00$	kN (满载) (Full Load)
	$P_{Wc2} = 0.00$	kN (空载) (No Load)
大车运行静阻力: Static resistance of trolley travelling:	$P_j = P_m + P_p + P_{WI}$	
	$P_{jc1} = 3.03$	kN (满载) (Full Load)
	$P_{jc2} = 2.15$	kN (空载) (No Load)

(2) 使用系数
(2) Use Coefficient

工作机最低工况系数: Minimum working condition coefficient of working machine:	$f_{Bmin} = 1.40$	(运行机构, 8h/天, Z=600 次) (Travelling mechanism, 8h/day, Z=600 times)
环境温度决定的服务系数: Service coefficient determined by ambient	$f_{BI} = 1.30$	(50°C) (50°C)

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temperature:

起动系数:
Starting coefficient: $f_{B1} = 0.93$

(负载持续率 40%)
(Continuous load rate: 40%)

使用系数:
Use coefficient:

$$f = f_B \min f_{B1} f_B$$

$$2 =$$

1.69

满足该使用系数!
It meets the use coefficient!

(3) 电动机选型

(3) Model Selection of Electric Motor

电动机台数:
Quantity of electric motor:

$m = 2$

(每个台车一套驱动机构, 一驱动轮, 一从动轮)
(Each set of trolley is designed with one set of drive mechanism, one drive wheel and one driven wheel)

运行机构总效率:
Total efficiency of travelling mechanism:

$\eta_{cz} = 0.95$

机构静功率:
Mechanism's static power:

$$P_N = \frac{P_{jc} \cdot v_{yc}}{1000 \eta_{cz} \cdot m} = 1.33 \text{ kW}$$

考虑 1.4 倍增大系数

1.86

kW

Considering the 1.4-fold increasing coefficient

型号: **SK2282ABGH**
Model: **SK2282ABGH**

P276

额定功率:
Rated power:

$P_n = 2.2$

kW

输出转速:
Output speed:

$n_n = 1450$

r/min

电动机输出轴的转矩:
Output shaft torque of electric motor:

$M_2 = 14.3$

N · m

机构的理论传动比:
Mechanism's the theoretical transmission ratio:

$i_0 = 22.78$

$(i_0 = n_n / n_{yt})$

实际传动比:
Actual transmission ratio:

$i = 21.9$

平均起动转矩倍数:
Average starting torque multiple:

$\lambda_{As} = 1.60$

平均起动转矩不可过大, 否则会发生打滑!
The average starting torque shall not be excessively large otherwise it will result in slipping!

最大起动转矩倍数:
Maximum starting torque multiple:

$\lambda_m = 2.40$

使用系数:
Use coefficient:

$f_B = 1.7$

减速器选择满足要求!
The reducer selection meets the requirements!

电动机输出轴轴径:
Output shaft diameter of electric motor:

$d_{d1} = 48$

mm

输出轴轴端长度:
Output shaft length:

$l_{d1} = 80$

mm

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输出轴许用转矩: $[T_n] = 941$ N · m
Allowable torque of output shaft:

自重: $m_1 = 69$ kg
Self-weight:

转动惯量: $J_1 = 0.0077$ kg · m²
Rotational inertia:

(4) 减速器输出转矩的校核
(4) Reducer Output Torque Check

运行阻力:
Travelling resistance:

$$F_{\max} = P_{m1} + P_{p1} + P_{W1} = 3.03 \text{ kN}$$

作用在减速器轴上的转矩:
Torque acting on reducer shaft:

$$T_{\max} = \frac{F_{\max} D_{yc}}{2m} = 190 \text{ N} \cdot \text{m}$$

减速器选择满足要求!
The reducer selection meets the requirements!

(5) 实际电动机转速
(5) Actual Speed of Electric Motor

转速校核公式: $n_N = \frac{v_{yt} i}{\pi D_{yt}}$
Speed check formula:

电动机实际转速: $n_{yc} = 1394$ m/min (满载)
Actual speed of electric motor: (Full Load)

(6) 过载校核
(6) Overload Check

$$P_n \geq \frac{1}{m \lambda_m} \left\{ \left[P_m + P_p + P_{WII} \right] \frac{v_{yc}}{1000 \eta_{cz}} + \frac{\sum J \cdot n^2}{91200 t_q} \right\}$$

$$\frac{1}{m \lambda_m} \left\{ \left[P_m + P_p + P_{WII} \right] \frac{v_{yc}}{1000 \eta_{cz}} + \frac{\sum J \cdot n^2}{91200 t_q} \right\} = 1.4 \text{ kW}$$

电动机满足过载要求!
The electric motor meets the overload requirements!

(7) 高速轴制动器 (电动机自带)
(7) High-speed Shaft Brake (Self-provided by Electric Motor)

$$M_z = \frac{1}{m} \left\{ \frac{(P_{WT} + P_p - P_m) D_{yc} \cdot \eta}{2i} + \frac{n \sum J}{9.55 t_z} \right\}$$

折算到电机轴上的总转动惯量: $\sum J' = 0.67$ kg · m²
Total rotational inertia converting to motor shaft: $\sum J' = 1.15m(J_1 + J_2) + \frac{(Q + TL + DL)D_0^2 \eta}{4i^2}$

制动器的制动力矩: $M_z = 6.2$ N · m
Braking torque of brake:

最大制动力矩: $M_{Z\max} = 20$ N · m
Maximum braking torque:

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允许制动力矩: $[M_z] = 6.2$ N · m
Allowable braking torque:

制动力矩满足要求!
The braking torque meets the requirements!

个数: $m_z = 4$ 台
Qty.: Unit

制动器选择满足要求!
The brake selection meets the requirements!

5. 机构起动时间和加速度的验算

5. Calculation of Mechanism Starting Time and Acceleration

① 起动时间

① Starting Time

$$t_q = \frac{n \sum J}{9.55 (m M_{dq} - M_{dj})}$$

电动机平均起动转矩: $M_{dq} = 22.9$ N · m
Average starting torque of electric motor:

电动机静阻力矩: $M_{dj} = 18.2$ N · m
Static torque of electric motor:

$$\left(M_{dj} = \frac{P_{jc} D_{yc}}{2000 i \eta} \right)$$

折算到电机轴上的总转动惯量: $\sum J = 0.738$ kg · m²
Total rotational inertia converting to motor shaft:

$$\sum J = 1.15 m (J_1 + J_2) + \frac{(Q + TL + DL) D_0^2}{4 i^2 \eta}$$

起动时间: $t_q = 4.0$ s
Starting time:

允许的起动时间: $[t_q] = 4.0$ s
Allowable starting time:

② 起动平均加速度

② Average starting acceleration

平均加速度: $a_q = 0.10$ m/s²
Average acceleration:

允许的起动平均加速度: $[a_q] = 0.21$ m/s²
Allowable average starting acceleration:

起动加速度满足要求!
The starting acceleration meets the requirements!

6. 运行打滑验算 (半驱动)

6. Check Calculation for Travelling Slipping (semi-driven)

小车离端梁中心的距离 (内侧): $S_d = 1150$ mm
Distance from hoist trolley to end beam center (inside):

移动载荷引起的轮压: $P_{Lc61} = 36.78$ kN (满载)
Wheel load caused by moving load: (Full Load)

(移动载荷在极限位置) $P_{Lc62} = 5.06$ kN
(Moving load is at the limit position)

$P_{Lc71} = 12.10$ kN (空载)
(No Load)

$P_{Lc72} = 2.42$ kN

满载最小轮压: $P_{Lcmin} = 20.98$ kN

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Minimum wheel load at full load:

空载最小轮压

Minimum wheel load at no load: $P_{Lmin} = 18.34$ kN

(1) 起动

(1) Starting

$$\left(\frac{\varphi}{K} + \frac{\mu \cdot d}{D}\right) P_{Lmin} \geq \frac{2000i \cdot \eta}{D} \left[M_{dq} - \frac{2000 \times \sum J \times i}{mD} a_q \right]$$

粘着安全系数:

$K = 1.20$

Adhesion safety coefficient:

粘着系数:

$\varphi = 0.120$

Adhesion coefficient:

$$\left(\frac{\varphi}{K} + \frac{\mu \cdot d}{D}\right) P_{Lmin} = 3.83$$
 kN

$$\frac{2000i \cdot \eta}{D} \left[M_{dq} - \frac{2000 \times \sum J \times i}{mD} a_q \right] = 3.82$$
 kN

起动运行不打滑!

The starting travelling is not slipping!

(2) 制动

(2) Braking

$$\left(\frac{\varphi}{K} - \frac{\mu \cdot d}{D}\right) P_{Lmin} \geq \frac{2000i}{\eta \cdot D} \left[M_z - \frac{2000 \times \sum J' \times i}{mD} a_z \right]$$

$$\left(\frac{\varphi}{K} - \frac{\mu \cdot d}{D}\right) P_{Lmin} = 3.50$$
 kN

$$\frac{2000i}{D \cdot \eta} \left[M_z - \frac{2000 \times \sum J' \times i}{mD} a_z \right] = 2.28$$
 kN

制动运行不打滑!

The braking travelling is not slipping!

7. 缓冲器

7. Buffer

(1) 选型

(1) Model Selection

$$W = \frac{m_{hc} v_c^2}{2} - P_{mt} S = E_t - P_{mt} S = \frac{1}{2} c S^2 \times n$$

最大减加速度:

$a_{max} = 4.0$

m/s²

Maximum deceleration and acceleration:

运行阻力:

$P_{mt} = 4.26$

kN

Travelling resistance:

碰撞瞬时动能:

$E_t = 1.9$

kN · m

Collision instantaneous kinetic energy:

碰撞质量:

$m_{hc} = 21.4$

t

Collision quality:

(满载运行)

(Full-load travelling)

碰撞前的速度折减系数: 50%

(配备限位开关)

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Velocity reduction coefficient before collision:

(Equipped with limit switch)

碰撞速度: $v_c = 0.42$ m/s
Collision speed:

单个缓冲器吸收的能量: $W = 0.8$ kJ/kN · m
Energy absorbed by single buffer:

大车缓冲器型号: **HCPU60-A**
Trolley buffer model: **HCPU60-A**

缓冲器容量: $[W] = 2.030$ kN · m
Buffer capacity:

容量满足要求!
The capacity meets the requirements!

缓冲器许用行程: $[S] = 148.0$ mm
Allowable buffer stroke:

行程满足要求!
The stroke meets the requirements!

最大撞击力: $P_e = 43$ kN
Maximum impact force:

弹性系数: $c = 0.1854$ kN/mm (弹性缓冲器)
Elastic coefficient: (Elastic buffer)

缓冲器效率: $\eta_h = 0.85$
Buffer coefficient:

单边个数: $n = 2.0$
Number of single edge:

单个重量: $m_{32} = 30$ kg
Single weight:

(2) 缓冲器末端力校核 (2) Buffer End Force Check

缓冲器的末端力:
End force of buffer:

$$P_{\max} = \frac{cS}{\eta} = 9.46 \text{ kN}$$

缓冲力满足要求!
The buffer force meets the requirements!
(弹性缓冲器)
(Elastic buffer)

签字:
Signature:
公司签章:
Company Stamp:

姓名: 彭军
Name: Peng Jun
日期: 2024 年 12 月 24 日
Date: December 24, 2024

职务: 机械工程师
Position: Mechanical Engineer